Global Bovaer Feed Ingredient Protocol – Dairy

Protocol ID	Version	Updated As Of
PRO-00000010	1.0	07/06/2026

Athian

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1 Introduction

This protocol provides a standardized, globally applicable methodology for quantifying greenhouse gas (GHG) emission reductions from the use of 3-nitrooxypropanol (3-NOP, marketed as Bovaer®) in dairy operations. Building on the Athian Bovaer® Protocol for US Dairy, it expands applicability across diverse geographies and data environments. The protocol may be applied in any jurisdiction where Bovaer® has received regulatory approval from the relevant authority and is intended for project developers registering projects with Athian across insetting programs or compliance frameworks by industry offtake partners working with Athian.

Livestock supply chains account for approximately 12% of global anthropogenic GHG emissions (FAO,2023). Enteric methane is a major contributor to dairy operation emissions and has a global warming potential (GWP_{100}) of 27.2 (IPCC, 2021). Strategies to reduce enteric methane are critical to achieving supply chain emission reduction goals. Feeding Bovaer® reduces enteric methane emissions by inhibiting the activity of the enzyme methyl-coenzyme M reductase (MCR), which catalyzes the final step in methane formation by rumen methanogenic archaea. This inhibition temporarily suppresses methane production during feeding. A body of peer-reviewed literature shows that Bovaer® reduces enteric methane production in dairy cattle on average by 30% (Kebreab et al., 2023), depending on dose and diet composition, which can be modeled to provide a more accurate reduction estimate.

The protocol provides guidance on quantifying GHG reductions associated with the use of Bovaer® in dairy cattle. Peer-reviewed Bovaer® data are used to model dose- and diet-dependent effects, enabling robust estimation of methane reductions under different production conditions. Animal productivity from feeding Bovaer® is not being impacted. Therefore, accurate monitoring, reporting, and verification (MRV) of its application is essential to ensure the credibility of quantified emission reductions and to support their use in carbon crediting, value chain interventions, or Scope 3 reporting.

Baseline emissions are calculated using IPCC (2019) Tier 2 for dairy cattle, reflecting animal performance and diet-specific parameters. Project emissions are determined by applying an enteric methane adjustment factor (AF), derived from peer-reviewed, dairy-specific dose–diet response models for Bovaer®. Upstream emissions associated with Bovaer® production and transport are included in the project boundary.

2 Project Definition

The intervention consists of the daily feeding of Bovaer® to dairy cattle in total or partial mixed rations feeding systems, in accordance with applicable label instructions and regulatory approvals in the project jurisdiction. All other husbandry practices are assumed to remain unchanged.

2.1 Impact on Yield

There is no impact of Bovaer® on dry matter intake, digestibility, or energy corrected milk. This is evidenced by several meta-analyses which are summarized in Table 2.1

Table 2.1 *Impact of Bovaer® on methane, dry matter intake, digestibility, and performance*

	Methane	Dry Matter Intake	Milk Yield	Adjusted Milk Yield[1]	Milk Fat	Milk Protein	Digestibility[2]
Jayanegara et al, 2018	-19%	=	=	=	Yield: NR %: ↑	Yield: NR %: =	=
Dijkstra et al., 2018	-39%	NR	NR	NR	NR	NR	NR
Kim et al., 2020	Linear ↓	=	=	=	Yield: NR %: =	Yield: NR %: =	=
Hristov et al., 2022	-28%	=	=	=	Yield: NR %: ↑	Yield: = %: =	NR
Kebreab et al., 2023	-32%	NR	NR	NR	NR	NR	NR

NR = not reported
[1]Adjusted for components as either energy-corrected milk, fat-corrected milk, or solids-corrected milk
[2]Includes dry matter, organic matter, and neutral detergent fiber (NDF) digestibility

2.2 Causality

Emissions reductions achieved through the use of Bovaer® are permanent and irreversible, presenting a meaningful opportunity to advance GHG mitigation efforts across agricultural supply chains. Without Bovaer®, methane emissions per cow are approximately 30% higher (Jayanegara et al., 2018; Dijkstra et al., 2018; Kim et al., 2020; Hristov et al., 2022; Kebreab et al., 2023). The emissions reductions from Bovaer® are permanent, creating a significant opportunity for GHG reductions via a financial incentive from the supply chain. Currently, dairy producers in many parts of the world receive no financial benefit from using Bovaer®. Participating in a carbon reduction incentive system, e.g., inset or offset market, provides an incentive through financial support for using Bovaer®. Participation additionally generates essential data streams that supply chain stakeholders can leverage to credibly report and deliver on global climate commitments.

3 Eligibility

Operations must feed Bovaer® to lactating dairy cattle according to the eligibility criteria below. Eligibility criteria are set at the intervention level, but additional specificity may be applied at the individual geographic level.

Criterion	Requirement
Directions	Bovaer® must be fed in accordance with the manufacturer's approved label instructions in the relevant jurisdiction.

Dose	60–90 mg of 3-NOP per kg of total ration dry matter (DM), unless a different dosage is approved or permitted by the relevant jurisdiction within the range noted in Section 3.1
Animal Types	Lactating dairy cattle; Mature dairy cows producing milk commercially. Excludes dry cows, heifers, calves, bulls, cull cows, non-lactating cattle, and multi-purpose cattle.
Location	Any country where Bovaer® is legally approved for use (for U.S. or U.S. tribal lands, the PRO-00000002 can be used)
Reporting Period	No minimum; maximum 12 months per reporting period. After 12 months. A project may continue, but it must use the most recent version of this protocol
Compliance	Attestation of voluntary compliance under applicable jurisdictional laws
Dry Matter Intake (DMI)	Dry matter intake shall remain between 18.2 kg DM/head/day and 28.0 kg DM/head/day
Neutral Detergent Fiber (NDF)	Neutral detergent fiber shall remain between 26.5% of dry matter and 43.5%
Crude Fat (CF)	Dietary crude fat shall remain between 2.8 % of dry matter and 5.8%

3.1 Location Specific Criteria

3.1.1 Australia Specific Eligibility Criteria

In Australia, methane inhibitors such as 3-nitrooxypropanol (3-NOP) are currently outside the regulatory scope of the Agricultural and Veterinary Chemicals Code (AgVet Code). Based on the Australian Pesticides and Veterinary Medicines Authority (APVMA), reduction of enteric methane is not considered a disease, condition, or productivity-related physiological modification as defined under Section 5(2) of the AgVet Code. As such, products that make methane-reduction claims are not classified as veterinary chemical products and do not require registration under the current regulatory framework. Therefore, Australian projects will use the Intervention Level Eligibility Criteria without adjustment.

3.1.2 Brazil Specific Eligibility Criteria

In Brazil, Bovaer® is approved for use in dairy cattle under the authorization of the Ministry of Agriculture, Livestock, and Supply (MAPA).The labeling requirement specifies a minimum inclusion rate of 600 g of Bovaer® 10 per metric ton of dry matter, which corresponds to approximately 60 mg of 3-NOP per kg of dry-matter intake. Bovaer® must be incorporated via approved pathways (e.g. premixes, mineral feed, or total mixed ration), and only use within this approved dosage and delivery framework is eligible for compliant labeling and claims.

3.1.3 Criteria for Adding New Geographies

- Regulatory: new geographies must demonstrate either regulatory approval of the use of Bovaer via label/dosage requirement information OR the upcoming approval of the use of Bovaer via documentation of submission for regulatory approval. If the latter, approval for addition of the new geography shall be contingent upon the approval of Bovaer for use in that geography and the provision of the label and dosage requirements.
- Traceability: new geographies must demonstrate sufficient measures for supply shed traceability.

- Monitoring plan: prior to the implementation of this protocol in a new geography, a new geography specific monitoring plan must be developed that accounts for the geography specific constraints identified in the eligibility section.
- Legal: prior to the implementation of this protocol in a new geography, the following legal requirements must be assessed:
  - The current state of any mandatory programs utilizing Bovaer.
  - Any geography specific attestation language that may differ from the default voluntary and legal compliance attestations used by Athian.
  - Any geography specific data privacy standards that may require adjustment to the Athian Data Quality Management Plan.
- Scientific Review: prior to the implementation of this protocol in a new geography, the applicability of Tier 2 region-specific parameters within the quantification must be reviewed and presented to the Athian Scientific Advisory Board for applicability to the new geography.
  - This review should also include a justification of the baseline scenario within that geography and the assessment of any regional adaptations for any regional systems.
  - For each new geography, a country inclusion assessment will be conducted to evaluate expected DMI, NDF, Crude Fat, and 3-NOP dose values against the validated prediction-model boundaries. This assessment will review available information for the geography of interest and document any expected parameter values that fall outside the validated model range. Country inclusion and country-specific representativeness checks do not modify, expand, or override the global prediction-model applicability limits

3.2 Voluntary Compliance & Performance Standard

All projects must demonstrate that the GHG emission reductions achieved through the intervention are not mandated by any applicable national, regional, or local laws or regulations. This includes, but is not limited to, requirements related to environmental protection, animal welfare, labor, safety, and biodiversity.

To demonstrate compliance, each project participant must sign an attestation of voluntary implementation prior to project start, confirming that the use of Bovaer® is not legally required. The Monitoring Plan must also include procedures for periodic review of applicable legal requirements at the project location to ensure continued compliance. Any changes in regulatory status that affect eligibility must be identified and addressed promptly.

3.3 Project Start Date

The start date for this intervention is no earlier January 1, 2026. The start date is defined as the first date that Bovaer® is incorporated into the lactating dairy cattle ration under this project.

3.4 Reporting Period

The reporting period is the period of time during which the intervention was implemented. Impact units or reductions for each project can be quantified as frequently as monthly (calendar month). The duration of this intervention is capped at a maximum of 5 years, or another determined at validation, when re-validation is required. Athian requires program review every 12 months and therefore re-validation may occur sooner than 5 years. Bovaer® itself is intended for continuous use by program participants for the duration of their participation.

3.5 Location

This intervention is designed to be globally applicable at the quantification level. However, there are considerations that must be made for baseline definitions, data reporting requirements, and other geographically-based legal and voluntary compliance standards. The geographic boundary of this intervention may be adapted as economic opportunities expand. The initial geographic boundaries are the countries of Australia and Brazil. In the U.S. or on U.S. tribal lands, the U.S. Bovaer protocol (PRO-00000002) can be used.

4 GHG Assessment Boundary

Project boundary for sources, sinks, and reservoirs was determined based on VM0041 methodology for determining project boundaries. More specifically: "The spatial extent of the project boundary encompasses all geographic locations of ingredient production, ingredient transport, and project activity locations where feed ingredient is part of the livestock production operation. No sinks are included in this program, as it solely focuses on reduction, not removals. The greenhouse gases (GHGs) included in, or excluded from, the project boundary are shown in Table 4.1

Table 4.1 *Description of all sources, sinks, and reservoirs evaluated for the protocol*

	Sources, Sinks, and Reservoirs	GHG	Included or Excluded	Justification
Baseline and Project	Direct Land Use	CO_2	Excluded	Emissions from land use do not change between the baseline and project scenarios.
	Enteric Fermentation	CH_4	Included	CH4 emissions from enteric fermentation are the major source of emissions in the project scenario.
	Fuel & Electricity Use	CO_2, N_2O, CH_4	Excluded	Emissions from electricity and stationary fuel use do not change between the baseline and project scenarios
	Ingredient Production & Transport	CO_2, N_2O, CH_4	Included	Emissions from the manufacturing and transportation of the feed additive intervention are included in this protocol.
	Manure Management	CH_4, N_2O	Excluded	Emissions from the management of manure do not change between the baseline and project scenarios.
	Waste Processing	CO_2, CH_4, N_2O	Excluded	Emissions from the management of dead animals do not change between the baseline and project scenarios.

5 GHG Quantification

For Bovaer®, baseline emissions are calculated according to Equation 5.3 in 2019 Refinement to the 2006 IPCC Guidelines for National GHG Inventories. Project emissions are calculated by determining the enteric methane emissions during the reporting period, adjusting those emissions in accordance with the Dose + NDF equation in Kebreab et al (2023), and adding emissions from the manufacturing and transport of Bovaer®. Consequently, emissions reductions are calculated by subtracting project emissions from baseline emissions, as follows:

$$ER_{Enteric} = BE_{Enteric} - PE_{Enteric} \quad \text{(Equation 5.1)}$$

Where:

$ER_{Enteric}$	=	Enteric emissions reductions from dairy cattle during the reporting period (MT CO_2e)
$BE_{Enteric}$	=	Baseline emissions from dairy cattle during the reporting period (MT CO_2e)
$PE_{Enteric}$	=	Project emissions from dairy cattle during the reporting period (MT CO_2e)

5.1 Baseline GHG Emissions

Baseline GHG emissions from enteric fermentation follow the Intergovernmental Panel on Climate Change (IPCC, 2019) Tier 2 methodology. Emissions for the baseline scenario are calculated according to the following:

$$BE_{Enteric} = CH_4 \times \frac{GWP}{1000} \quad \text{(Equation 5.2)}$$

Where:

$BE_{Enteric}$	=	Baseline emissions from lactating dairy cattle during the reporting period (MT CO_2e)
CH_4	=	Baseline methane emissions for lactating dairy cattle during the reporting period (kg CH_4)
GWP	=	Global warming potential of methane over a 100-year time frame; 27.2 is used per IPCC, 2021 (kg CO_2e per kg CH_4)
1000	=	kg CO_2e per MT CO_2e

Baseline methane emissions for lactating dairy cattle during the reporting period are calculated according to the following:

$$CH4 = \frac{(DMI \times GE) \times \left(\frac{Ym}{100}\right) \times C \times t}{55.65} \quad \text{(Equation 5.3)}$$

Where:

CH_4	=	Baseline enteric methane emissions for lactating dairy cattle during the reporting period (kg CH_4)

DMI	=	Average daily dry matter intake for lactating dairy cattle during the reporting period (kg DM per head per day)
GE	=	Gross energy concentration of lactating dairy cattle diet (MJ per kg dry matter)
Y_m	=	Methane conversion factor for lactating dairy cattle during the reporting period (percentage of dietary gross energy lost as CH_4; obtained from **Table 5.1** Methane Conversion Factors (Ym) for lactating dairy cattle from Table 10.12 of IPCC, 2019 adapted from Table 10.12 of IPCC, 2019)
C	=	Average number of total lactating dairy cattle during the reporting period (head)
t	=	Total number of days in the reporting period (days)
55.65	=	Energy content of methane (MJ per kg CH_4)

The appropriate Ym is selected from Table 5.1 based on diet nutrient composition. Nutrient composition reported as the weighted averages based on feed ingredients delivered, all documented in the Athian platform.

Table 5.1 *Methane Conversion Factors (Ym) for lactating dairy cattle from Table 10.12 of IPCC, 2019*

Lactating dairy cows diet characteristics	Ym (%)	Notes
DE ≥ 70%, NDF ≤ 35%	5.7	High digestibility, low fiber diet
DE ≥ 70%, NDF > 35%	6.0	High digestibility, moderate-high fiber diet
DE 63–70%, NDF > 37%	6.3	Moderate digestibility, higher fiber diet
DE ≤ 62%, NDF > 38%	6.5	Low digestibility, high fiber diet

DE= Diet digestible energy (%)
NDF= Diet neutral detergent fiber (%)

Note: Where diet characteristics, DE, or NDF fall outside the defined ranges (e.g., a diet with 69% DE and 36% NDF), a conservative Ym value shall be selected and justified. If no Ym value is available, 5.85 is recommended, as it is the average of the two tiers covering NDF ≤ 37%. This is accurate because this selection follows NDF constraints (≤37% NDF) more than DE constraints, which agrees with other scientific equations and opinions that NDF is a major factor driving methane production (NASEM, 2021). The approach is also conservative because by following NDF constraints more closely than DE, a lower Ym is used, which results in smaller estimates of baseline methane and thus smaller estimates of carbon reductions due to the implementation of Bovaer®.

Where diet digestible energy is not available to producers to report directly, producers may choose to use the ***Athian Dairy Total Digestible Nutrients (TDN) Methodology****. This requires the producer reporting of the following additional variables: Dietary Crude Protein (CP, %), Dietary Ash (%).*

5.2 Project GHG Emissions

Emissions in the project scenario are estimated as the sum of emissions during the reporting period from enteric fermentation and the production and transport of Bovaer® according to the following equation:

$$\mathrm{PE}_{Enteric} = BE_{Enteric} \times \left(1 - \frac{|AF_H|}{100}\right) + GHG_{MT} \qquad \text{(Equation 5.4)}$$

Where:

$PE_{Enteric}$	=	Total project enteric CH4 emissions from dairy cattle during the reporting period (MT CO_2e)
$BE_{Enteric}$	=	Baseline emissions from dairy cattle during the reporting period (MT CO_2e)
AF_H	=	Group enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer®; **Equation 5.8a** or **Equation 5.8b**)
GHG_{MT}	=	Emissions associated with Bovaer® manufacturing and transport during the reporting period (MT CO_2e)

5.2.1 Enteric Methane Emissions Adjustment Factor

The total lactating dairy cow herd enteric methane adjustment factor is determined by the following equations. AF_H shall be set to 0 if the reported dose falls outside of the parameters detailed in the Dose Variation Guidelines in the monitoring plan. If NDF, DMI, or CF fall outside the model range, the procedures outlined in Section 3 should be followed.

$$AF_H = AF \times PBCD \times GEF \qquad \text{(Equation 5.5)}$$

Where:

AF_H	=	Group enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer®)
AF	=	Enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer®; **Equation 5.8a** or **Equation 5.8b**)
$PBCD$	=	Percent Bovaer®-fed cattle-days of total cattle-days available (%; **(Equation** 5.6))
GEF	=	Average grazing exposure factor

Percent Bovaer®-fed cow-days is calculated according to the following equation:

$$PBCD = \frac{\sum_i BC_i \times Bt_i}{C \times t} \qquad \text{(Equation 5.6)}$$

Where:

$PBCD$	=	Percent Bovaer®-fed cow-days of total cow-days available (%)
BC_i	=	Average number of lactating dairy cattle in group i fed Bovaer® during the reporting period (head)
Bt_i	=	Number of days in the reporting period Bovaer® was fed to group i (days)
C	=	Average number of total lactating dairy cattle fed during the reporting period (head). If multiple groups of cattle included in one project, this is the sum of the average number of each group.

t	=	Total number of days in the reporting period (days)
i	=	Different groups of Bovaer®-fed lactating dairy cattle where days and animal numbers differ between groups

The use of PBCD as a measure of the available days of effect of Bovaer on the animals in question provides conservativeness and consistency in reporting to cover non-delivery gaps, which would not be covered in the number of days in the reporting period that Bovaer was fed to group i.

In systems where animals are partially grazed, the effectiveness of Bovaer® depends on the duration of exposure via the total mixed ration (TMR). Based on experimental evidence (de Gucht et al., 2025, partial grazing (e.g., up to 6 hours per day) does not reduce methane mitigation efficacy compared to full indoor feeding systems.

To account for this, a Grazing Exposure Factor (GEF) is applied:

$$GEF = \begin{cases} 1, \; if \; H_g \leq H_c \\ \frac{24 - H_g}{24 - H_c}, \; if \; H_g > 0 \end{cases} \qquad \textit{(Equation 5.7)}$$

Where:

GEF	=	Average grazing exposure factor
H_g	=	Average grazing hours per day
H_c	=	Maximum grazing hours for which no reduction in methane mitigation efficacy is assumed

Kebreab et al. (2023) conducted a meta-analysis of 14 studies, including 48 treatment means, to determine the effect of Bovaer® (3-NOP) on enteric methane production in lactating dairy cattle. The analysis produced a Dose + NDF + Fat model that quantifies the methane reduction effect based on the 3-NOP dose (mg/kg diet DM), the diet's neutral detergent fiber (NDF, % DM), and dietary crude fat (% DM). This model (Equation 5.8a) is preferred because it incorporates an additional dietary variable—fat percentage—that influences enteric methane production, thereby improving the precision of the adjustment factor.

The enteric methane adjustment factor due to Bovaer® implementation is calculated according to the following:

$$\mathrm{AF} = -32.36 - 0.282 \text{ x } (\mathrm{DOSE} - 70.5) + 0.915 \text{ x } (\mathrm{NDF} - 32.9) + 3.08 \text{ x } (\mathrm{Fat} - 4.2) \qquad \textit{(Equation 5.8a)}$$

In cases where reliable dietary fat data is not available, a reduced model omitting the fat term may be used. This alternative (Equation 5.8b) still accounts for the primary drivers — dose and NDF:

$$AF = -32.8 - 0.285 \times (DOSE - 70.5) + 0.633 \times (NDF - 32.9)$$ (Equation 5.8b)

Where:

AF	=	Enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer®)
$DOSE$	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
NDF	=	Neutral detergent fiber content of the diet (% DM)
Fat		Crude fat content of the diet (% DM)

5.2.2 GHG Emissions from Bovaer® Manufacturing and Transport

Emissions from the feed ingredient are estimated by including all GHG sources from manufacturing and transport. Accounting for these GHG sources is not required for a project where such emissions are shown to be de minimis.[1] Otherwise, these emissions must be estimated as follows:

$$GHG_{MT} = GHG_M + GHG_T$$ (Equation 5.9)

Where:

GHG_{MT}	=	Emissions associated with Bovaer® manufacturing and transport during the reporting period (MT CO_2e)
GHG_M	=	Emissions associated with Bovaer® manufacturing during the reporting period (MT CO_2e)
GHG_T	=	Emissions associated with Bovaer® transport during the reporting period (MT CO_2e)

The total amount of product used during the reporting period is calculated according to the following:

$$Bovaer_{Product} = \frac{DMI \times DOSE}{AI \times 1000000} \times PBCD \times C \times t$$ (Equation 5.10)

Where:

$Bovaer_{product}$	=	Amount of Bovaer® product used during the reporting period (kg)
DMI	=	Average daily dry matter intake during the reporting period (kg DM per head per day)
$DOSE$	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
$PBCD$	=	Percent Bovaer®-fed cattle-days of total cattle-days available (%)

[1] The pool or source may be excluded only if it is determined to be insignificant using appropriate approved tools for significance testing (e.g., the CDM Tool for Testing Significance of GHG Emissions in A/R CDM Project Activities, v01. Available at http://cdm.unfccc.int/EB/031/eb31_repan16.pdf).

C	=	Average number of total dairy cattle fed during the reporting period (head)
t	=	Total number of days in the reporting period (days)
AI	=	Active ingredient content of Bovaer® product(kg 3-NOP per kg Bovaerproduct)
1000000	=	mg 3-NOP per kg 3-NOP

Manufacturing-based GHG emissions associated with Bovaer® production are supplied by the manufacturer based on ISO standards 14040.44:2006 and ISO 14067:2018 and calculated according to the following. The Product Carbon Footprint (PCF) of Bovaer® represents the GHG emissions associated with the production and manufacturing of Bovaer® within the defined PCF system boundary. Transport emissions included within the project boundary are quantified separately using the methodology described below.

$$GHG_M = \frac{Bovaer_{Product} \times PCF}{1000} \quad \text{(Equation 5.11)}$$

Where:

GHG_M		Emissions associated with Bovaer® manufacturing during the reporting period (MT CO_2e)
$Bovaer_{product}$	=	Amount of Bovaer® product used during the reporting period according to **(Equation** 5.10) (kg)
PCF	=	Product Carbon Footprint of Bovaer® product production (kg CO_2e per kg *Bovaer*$_{product}$)
1000	=	kg CO2e per MT CO_2e

Emissions from the transport of the feed ingredient to the project site are calculated as:

$$GHG_T = \frac{Bovaer_{Product} \times (EFT_R + EFT_O)}{1000} \quad \text{(Equation 5.12)}$$

Where:

GHG_T	=	Emissions associated with Bovaer® transport during the reporting period (MT CO_2e)
$Bovaer_{product}$	=	Amount of Bovaer® product used during the reporting period according to **(Equation** 5.10) (kg)
EFT_R	=	Emissions factor from road transportation of Bovaer® product during the reporting period (kg CO_2e per kg *Bovaer*$_{product}$)
EFT_o	=	Emissions factor from ocean transportation of Bovaer® product during the reporting period (kg CO_2e per kg *Bovaer*$_{product}$)
1000	=	kg CO_2e per MT CO_2e

Emissions from the transportation of Bovaer® are estimated using internationally recognized emission factor databases for freight transport, including road and ocean transport. Transport distances shall be determined based on representative supply chain pathways between production facilities, distribution centers, and project locations. Where project-specific transport data are unavailable, conservative default distances and transport modes shall be applied. Default transport distances and modes are defined as standardized, conservative assumptions intended to reflect typical regional (road) and international (ocean) supply chains for feed additives. For example, a project may assume regional road transport from a distribution center to the farm and international ocean freight from manufacturing location to the destination country.

5.3 Fat and Protein Corrected Milk (FPCM)

The good impacted by this intervention is fat and protein corrected milk (FPCM). This is quantified as follows:

$$FPCM = Milk\ Shipped\ x\ [0.1226\ x\ fat\ (\%)\ x\ 0.0776\ x\ protein\ (\%) + 0.2534] \qquad \text{(Equation 5.13)}$$

Where:

$FPCM$	=	Fat and Protein Corrected Milk from lactating dairy cattle during the reporting period (kg)
$Milk\ Shipped$	=	Quantity of milk shipped (kg) to processor during the reporting period
0.1226	=	Fat coefficient
fat	=	% fat shipped to processor during the reporting period
0.0776	=	Protein coefficient
$protein$	=	% true protein shipped to processor during the reporting period. If true protein is not available, it may be estimated using crude protein shipped minus 0.19
0.2534	=	Intercept

5.4 Leakage & Permanence

Bovaer® is metabolized by its own mode of action (Duin et al., 2016) and is not practically excreted in the environment (Bampidis et al., 2021). Studies between control and 3-NOP-fed animals have shown no differences in fecal characteristics (volatile solids, carbon, nitrogen or total solids) in forage- or grain-based diets (Nkemka et al., 2019), no differences in manure GHG and NH3 emissions or manure application to crop yield (Owens et al., 2020, 2021), and no impact on anaerobic digestion (Nkemka et al., 2019). Therefore, because Bovaer® does not affect any GHG emission sources outside the identified GHG boundaries, leakage is not relevant to this project and no deductions will be applied to outcomes generated according to this protocol. This protocol assumes no effect and thus is conservative relative to a scenario in which there could be a decrease in dry matter intake.

In the context of this protocol, leakage could also involve changes in livestock populations resulting from feed ingredient impacts on performance. Bovaer® does not influence productivity or other greenhouse gas (GHG) emission sources outside the established boundaries of the project. As a result, leakage is considered zero, and no deductions will be applied to credits generated under this program. Bovaer® acts by reducing enteric methane;

thus, the reductions in GHG emissions associated with feeding Bovaer® are permanent and irreversible.

5.5 Uncertainty

The major risk that could substantially affect the intervention's GHG emission reductions is in the estimation of feed intake. Traditionally, a large amount of uncertainty in dairy operation greenhouse gas calculations originates from estimations of feed intake and digestibility, which vary widely between and within cattle and geographies. A significant portion of this uncertainty is mitigated in this program through utilizing each operation's individual feed ingredient data to quantify dry matter intake and use contemporaneous estimates of gross and digestible energy for the quantification of both baseline and intervention emissions. The availability of these data facilitates a precise measure of gross energy intake from which enteric emissions reductions are calculated. Because feed intake is recorded and received directly from the dairy farm, the remaining uncertainty associated with the GHG reductions quantified in this program arise from the adjustments made around the intervention of Bovaer® implementation. These adjustments are taken directly from a peer-reviewed meta-analysis. Bovaer® has reliable and substantiated data regarding its mode of action and effect, and a summary of reported error associated with these effects is reported in Kebreab et al. (2023)

Because uncertainty associated with feed intake, a primary driver in GHG emissions, was mitigated through incorporating operation-specific feed data in a Tier 2 approach to calculate GHG emissions and uncertainty associated with adjustment factors for Bovaer®'s effect were mitigated using a meta-analysis, no deductions for uncertainty will be taken.

5.6 Deviations from Protocol Methodologies

Deviations from the methodologies in section 5 of this protocol are not allowed. Any deviation must be documented and justified in the monitoring report.

6 Monitoring

This program is 100% monitoring. All producers participating in the program will go through verification of their baseline data and verification of monitoring periods. A monitoring plan has been developed for all monitoring and reporting activities associated with the project, standardized across all participating farms.

Verifiers will use the monitoring plan and report to confirm that the requirements of this program have been met. This monitoring plan provides the processes, requirements, and sources of information necessary to assess the GHG reductions created by the practices included in this protocol.

This includes:

1. The procedures for collecting data on intervention activities related to implementation.
2. The data points collected to verify emission reduction, project, and baseline calculations.

3. The QC/QA processes to ensure the accuracy and consistency of the data collected.

The monitoring reports described in the monitoring plan include the following elements:

1. General description of the project, including the location of the cattle operations
2. List of the practices implemented
3. Description of the process and frequency of data collection and the archiving procedures
4. Recordkeeping plan
5. Role of any individuals performing activities related to the practices implemented
6. Quality assurance/quality control (QA/QC) procedures to ensure the accurate collection and entry of data in quantification systems
7. Monitoring reports must include the monitoring time period.
8. Monitoring reports must include the list of parameters measured and monitored.
9. Monitoring reports must include the types of data and information reported, including units of measurement.
10. Monitoring reports must include the origin of the data.
11. The monitoring report must include an attestation as to regulatory compliance.
12. The monitoring report should be submitted no less frequently than annually and no more frequently than 30 days.
13. The monitoring period can be as short as 30 days. The maximum monitoring period is 12 months.
14. The monitoring report must be submitted and shared with Athian, as the program administrator.

A monitoring period represents a calendar month. Where necessary, multiple months may be combined at the discretion of the verifier. When the monitoring period is over (i.e. the month has fully passed), data from operating records are input into the Athian quantification tool to assess the impact of the intervention for that monitoring period. Supporting documents are collected concurrently. Once all required inputs and supporting documentation are collected, a third-party verifier receives the information to assess the validity of the reported inputs as well as verify the quantification themself. Participating producers are given the opportunity to produce further documentation of any values the verifier determines to be non-compliant before a final decision on the results of the monitoring period is rendered, either verified or not.

This monitoring plan provides the requirements and sources of information necessary to assess the GHG reductions created using this protocol. This monitoring plan describes the procedures for collecting data on intervention activities related to the implementation of protocol practices. The data collected will support emission reduction and baseline calculations. This monitoring plan also outlines the QC/QA processes to ensure the accuracy and consistency of the data collected that will be used to verify emissions reduction outcomes.

6.1 Data Quality Assurance

The Athian Data Quality Management Plan aims to ensure that a producer's data is accurate, reliable, and fit for its intended purpose to assess the impact of the practices included in this protocol on CO_2e emissions associated with the management of manure and feed cultivation. The goals and objectives of a can be categorized into several key areas, each targeting different aspects of data quality management. These include accuracy, timeliness, comparability, and creditability.

Accuracy:

- Data Collection Methods: Data will be provided by the farm directly based on various on farm systems.
- Consistency Checks: Input forms will check for data type and range preventing grossly invalid data from being entered.
- Method Validation: Based on type, input may be limited to certain ranges or values. Additionally, producers must attest to and confirm accuracy.

Timeliness:

- Data Collection Frequency: As defined by the protocol
- Data Reporting Schedule: Specific schedules are defined by the protocol monitoring plan.
- Response Procedures for Data Variations: Significant data issues should be prevented at entry. Additionally identified issues can be corrected by Athian staff as needed.

Comparability:

- Standardized Methods Used: Form input is used to collect quantitative data from on farm systems.
- Benchmarking: Per the Athian Data Retention Policy, all GHG related data is kept for a minimum of 10 years. Data, in aggregated and anonymized form, can be used for benchmarking if/when applicable.

Creditability:

- Documentation of Data Processes: All data processes are ultimately governed by the Athian Data Protection, Data Retention, and Software Development Lifecycle (SDLC) policies. These policies are maintained as controlled documents in the Athian compliance system (Drata) and are reviewed and updated at least annually.
- Transparency Measures: Data transparency is critical to credibility and integral to the data collection process. Data input directly in the platform from producers requires an attestation from the producer as to the accuracy before being submitted for verification. Data collected via an integration to a 3rd party data collection software also requires the producer to attest to the accuracy. In addition, producers have visibility as to the data provided to 3rd party verifiers and can see the status of the verification of each element of data submitted. All verification reports include each data element collected and reviewed as part of the verification process for complete transparency in the reporting of the emissions result.

The Athian platform has a comprehensive set of automated processes that confirm the integrity, correctness, and completeness of data. These include checking the data upon ingestion from any 3rd party data source, inclusive of data delivered via API or manually entered, for completeness and accuracy. These checks include verification of appropriate formatting, field-level requirements that ensure the presence of all required data, and identification of any data variance from the previously verified data. If errors are identified, notifications are generated and delivered to engineering, product management, and service management for resolution. Those parties then determine the source and scope of the issue(s), engage any necessary participating party, resolve and document the identified issues.

In addition to the data validation checks identified above, Athian has implemented a service-driven approach for applying logic consistently, significantly reducing the potential for error in the process. The programmatic logic used reduces or replaces much of the process that is prone to human error. The Athian platform hosts the mechanisms for documenting any data discrepancy as well as their respective severity and solution. The platform retains a complete transaction history for all data ingested, inclusive of date/time stamp and the individual user or software supplying the information. This ensures that Athian will have a complete history/picture of all data used when rendering a decision or result.

All data used to meet GHG carbon accounting standards for impact units must be retained for a minimum of ten years. This includes producer business contact information, location information, monitoring period information, and all verification information. All of the aforementioned processes and procedures adhere to industry best practices, including SOC 2 review. The quantification tool for this program is thoroughly tested against known results of data sets any time updates to the quantification methodology or tool are made. The tests follow the same methods as used for the Simulated Quantification Model and are checked against the quantification methodology for accuracy by the Athian development team.

6.2 Product Quality Assurance

Bovaer® shall be incorporated into dairy cattle rations under the supervision of qualified animal nutrition professionals or trained personnel, in accordance with the approved label instructions in the relevant jurisdiction.

Rations may be formulated using ration balancing tools or equivalent methods to ensure appropriate inclusion rates of Bovaer® alongside other dietary components (e.g., macronutrients, micronutrients, and feed ingredients). Where such tools are used, they shall rely on validated feed composition data and documented ingredient specifications.

Bovaer® may be incorporated into the ration through various delivery systems, including total mixed rations (TMR), premixes, or automated/micro-dosing systems, depending on the production system. Feed manufacturing, storage, handling, and delivery processes shall comply with applicable feed safety and quality standards in the jurisdiction of operation, including good manufacturing practices and risk-based controls for animal feed.

Project participants are responsible for ensuring that Bovaer® is applied at the correct dose and consistently delivered to the intended animal groups. Any deviations from label instructions or feeding protocols shall be documented and addressed through quality control procedures.

Responsibilities for ration formulation, feed preparation, and feeding practices may be shared among nutritionists, feed suppliers, and cattle operators; however, all parties shall ensure that implementation is consistent with protocol requirements and documented for verification purposes.

***Table 6.1** Monitoring parameters*

Data Parameter	Description	Unit	Data Source	Frequency	Values Applied	Methods/QA/QC	Responsible	Data Management
C	Average number of cattle during the reporting period	head	Farm operating records	Per reporting period	Calculated as the average inventory of cattle during the monitoring period	Reconciled with inventory, purchase, and sales records	Farm operator	Stored in quantification system
t	Length of reporting period	days	Monitoring plan/reporting records	Per reporting period	Number of days in reporting period	Cross-check against reporting dates	Project Developer	Recorded in monitoring report
BN_i	Average number of cattle in group (i) fed Bovaer®	head	Farm logs; farm operating records	Per reporting period	Recorded for each treated group	Cross-check against pen/group records	Farm operator	Stored in quantification system
Bt_i	Number of days Bovaer® was fed to group (i)	days	Feeding logs	Per reporting period	Recorded for each treated group	Cross-check against feeding period dates	Farm operator	Stored in quantification system
H_g	average grazing hours per day	hours	Feeding logs	Per reporting period	Recorded for each treated group	Cross-check against feeding period dates	Farm operator	Stored in quantification system
H_c	maximum grazing hours for which no reduction in methane mitigation efficacy is assumed	hours/day	Protocol constant based on ILVO experimental evidence (2025).	Fixed (reviewed only upon protocol revision).	6 hours/day	Fixed protocol parameter. Correct implementation verified during quantification and verification.	Project Developer	Maintained as a fixed parameter within the protocol and calculation tool.
DMI	Average daily dry matter intake per head	kg DM/head/ day	Feed logs; ration sheets; lab analysis	Per reporting period	Weighted average intake during reporting period	Mass balance checks; dry matter validation; audited at verification	Producer / Nutritionist	Ration history stored with supporting records
DE	Digestible energy of the diet	% of gross energy	Calculated from feed composition data	Per reporting period	Computed using primary feed data and recognized reference tables	Plausibility checks; documented calculation basis	Nutritionist / Project Developer	Inputs and equations version-controlled

NDF	Neutral detergent fiber content of the diet	% DM	Calculated from feed composition data	Per reporting period	Computed using ration formulation and feed composition data	Cross-checked against feed tables and ration sheets	Nutritionist / Project Developer	Stored in quantification system
Fat	fat content of the diet	% DM	Calculated from feed composition data	Per reporting period	Computed using ration formulation and feed composition data	Cross-checked against feed tables and ration sheets	Nutritionist / Project Developer	Stored in quantification system
GE	Gross energy concentration of diet	MJ/kg DM	Feed composition data or default value	Per reporting period	Calculated from diet composition; default may be used if feed-specific data are unavailable	Use recognized feed tables; plausibility checks	Project Developer / Nutritionist	Stored in quantification system
Y_m	Methane conversion factor selected for the diet	%	IPCC Tier 2 guidance or approved national defaults	Per reporting period	Selected based on DE and NDF	Verified against protocol table and diet class	Project Developer	Stored in quantification system
AF	Enteric methane Reduction	%	Model-derived using Equation 5.8a (Dose, NDF, Fat) or Equation 5.8b (Dose, NDF)	Per reporting period	Calculated from DOSE and NDF	Model validation; input verification	Project Developer	Stored in quantification system
$DOSE$	3-NOP dose in diet	mg 3-NOP/kg DM	Feeding records / ration formulation	Per reporting period	Label-compliant dose or approved program value	Verified against feed formulation and delivery records	Farm operator / Nutritionist	Stored in quantification system. Batch traceability maintained
AI	Active ingredient content of Bovaer® product	kg 3-NOP/kg product	Manufacturer specification / approved formulation	Per reporting period or when formulation changes	Product-specific value	Verified against product specification and formulation records	Project Developer / Nutritionist	Stored in quantification system
PCF	Product carbon footprint of Bovaer® product	kg CO_2e/kg product	Manufacturer LCA or approved program default	Updated periodically	Most recent approved value	Verified against LCA documentation	Project Developer	Stored in quantification system

EFT_R	Road transport emission factor for Bovaer® product	kg CO_2e/kg product	Approved secondary databases / route calculations	Per reporting period	Route-specific or conservative default	Cross-checked against recognized databases and assumptions	Project Developer	Stored in quantification system
EFT_o	Ocean transport emission factor for Bovaer® product	kg CO_2e/kg product	Approved secondary databases / route calculations	Per reporting period	Route-specific or conservative default	Cross-checked against recognized databases and assumptions	Project Developer	Stored in quantification system

7 Reporting

Project developers must provide the following documentation each reporting period to generate credits from this protocol:

1. Name and address of the project developer
2. List of all of the operations included in the project including the owner/operator contact information and address of the operation
3. Regulatory compliance documentation and attestation
4. Monitoring plan
5. Monitoring report with all the data used in the calculations for Section 5 of the protocol
6. Monitoring report must include the intended use and user of the monitoring report.

7.1 Record Keeping

For purposes of third-party verification and historical documentation, project developers must keep all information listed in this protocol for a period of 10 years after the information is generated. The information the project developer should retain includes:

1. All data inputs for the calculation of the project emission reductions as well as the results of emission reduction calculations
2. Copies of all permits, Notices of Violations (NOVs), and any relevant administrative or legal orders dating back at least 3 years prior to the project start date
3. All verification records and results
4. All maintenance records relevant to the monitoring equipment

8 Verification

Verification bodies will contract directly with Athian for all validation and verification engagements.

Projects verified under this protocol will meet, at minimum, the auditing standard of limited assurance and adhere to 14064-3. The verification body must provide a factual statement expressing the outcome of the verification.

Issues identified during verification must be classified by verification bodies as either material (significant) or immaterial (insignificant). To be verified successfully, all reported emissions reductions must be free of material misstatements.

All projects developed under this protocol must achieve >95 percent level of accuracy. This means that the project's calculated emission reductions must be less than 5 percent different than those calculated by the verifier.

8.1 Verification Body Requirements

To conduct verification under this protocol, all Validation and Verification Bodies (VVB) must meet the following criteria:

1. Accreditation under International Organization for Standardization (ISO) 14065: 2013 with conformance to all accreditation requirements under ISO 14065, ISO 14064-3: 2006, IAF MD 6: 2014 and all other accreditation requirements, or Acceptance in the American National Standards Institute (ANSI) accreditation program, having filed a full application for ISO 14065: 2020
2. Demonstrated/documented subject matter expertise in the on-farm operations related to an approved protocol (e.g., Dairy Operations; Feed Lot Operations)
3. Demonstrated/documented experience in a particular region or state where the verification will occur
4. Monitoring conducted in accordance with the requirements of the relevant protocol
5. Monitoring conducted in a manner that allows for a complete and transparent quantification of GHG reductions

8.2 Conflict of Interest

When conducting verification under this protocol verifiers must be seen as credible, independent, and transparent. To meet this requirement, a conflict of interest (COI) determination must be made prior to starting any verification activities. A COI occurs in any situation that compromises the verifier's ability to perform an independent verification. Every verifier must provide information about its organizational relationships, internal structures, and management systems for identifying potential COIs. Verifiers must evaluate any potential conflicting services it has provided to the project developer, including any advice or consulting provided outside of the verification process.

8.3 Verification Process

To verify the project, the verifier must develop a risk-based verification plan that considers the size and complexity of the project and the relevant sector, technology, and processes. The verifier must follow the following process:

1. Complete a COI evaluation. If there is a potential COI, the verifier is not allowed to conduct the verification.
2. Prepare a verification plan that includes, at a minimum:
    a. A list of people from the VVB involved in the verification
    b. A list of the location and dates of any on-site visits that will be conducted
    c. The types of data and documents that will be reviewed by the verifier
    d. A list of the people who are expected to be interviewed as a part of the verification
3. Conduct a kick-off meeting with all parties to lay out the timeline and process of the verification.
4. Conduct, at minimum, one annual on-site visit to confirm practice implementation

5. Undertake a desk review of the data from the project.
6. Prepare a verification report that includes:
    a. A verification statement documenting the outcome of the assessment (reduction results) and if there was any material discrepancy noted
    b. Key details about the project including: producer and farm operation identification, verifying body and lead verifier contact information, protocol information, and intervention information
    c. A description of the protocol, the objectives and criteria used to arrive at the final result, the scope of the project, the level of assurance associated with the project, and any details about the implementation of the practices observed
    d. Detail about the verification process used to complete the assessment including approach and methods and also noting any conflict of interest
    e. Verification findings including confirmation of producer eligibility, adherence to the criteria established in the protocol, the verified emissions quantification values, and the final written opinion of the verifier(s)
    f. An issue log capturing any issues identified during the verification and their classification as either material (significant) or immaterial (insignificant)
    g. A representation of all data / documents used in the process of verification

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